

Reading Note for Slide 2

Data analysis snapshot | Meds Counsulting LLC

Opening

This slide summarizes our initial data analysis for NHN. The purpose is to show what we checked before moving into deeper modeling: missing data, outliers, trend patterns, and early correlations with 30-day readmission.

1. Data foundation

We analyzed three cleaned datasets: patient readmission, financial impact, and care coordination. Together, they cover 12,000 records and 12,000 unique patients. The overall 30-day readmission rate in the cleaned patient-level file is 43.2%.

The first quality signal is that 30.2% of rows have at least one data quality flag. That does not mean the data is unusable, but it tells us to be careful when interpreting follow-up, cost, and outlier-related results.

2. Missing data

The largest missing data issue is follow-up status. It is missing for 2,587 patients, or 21.6 percent of the dataset. This matters because follow-up status is directly related to the care transition process we are trying to evaluate.

Other missing fields are smaller, mostly around 1.5 to 2.0 percent. These include insurance type, length of stay, penalty cost, follow-up program cost, and follow-up timing. The main takeaway is that follow-up status needs special attention before we use it in any final model or recommendation.

Missing field	Rows	Rate
follow-up status	2,587	21.6%
insurance type	240	2.0%
length of stay	240	2.0%
penalty cost	240	2.0%
follow-up program cost	240	2.0%

Transition cue

After explaining missing data, move to outliers and quality flags as the second part of data readiness.

3. Outliers and quality flags

For outliers, we used the IQR rule to identify unusually high or low values. Length of stay has 191 outlier records, with values above 11.5 days flagged. Total care cost has 72 outlier records above about 41,217 dollars. Net reimbursement gap has 64 outlier records above about 30,691 dollars.

We also found a small number of negative financial values, including 49 negative penalty-cost records. These are small as a share of the full dataset, but they should be reviewed because financial values can affect ROI and cost-savings estimates.

4. Trend pattern

The clearest trend on the slide is by age group. Readmission increases from 42.0 percent for patients aged 50 to 64 to 54.3 percent for patients aged 85 and older. This suggests that older patients may need more targeted discharge planning and post-discharge support.

This trend does not prove that age alone causes readmission. Instead, age should be treated as a risk signal that may interact with chronic conditions, prior admissions, length of stay, and follow-up completion.

5. Correlation signals

The first correlation check shows that prior admissions and chronic conditions have the strongest positive relationship with readmission. Prior admissions has a correlation of 0.22, and chronic conditions has a correlation of 0.21. Length of stay and age are also positively related, but slightly weaker.

These are early signals, not final conclusions. Correlation helps us decide which variables deserve more attention in the next stage, but we still need predictive modeling and segment analysis to understand which factors matter most after controlling for overlap.

Variable	Correlation
Prior admissions	0.22
Chronic conditions	0.21
Length of stay	0.15
Age	0.14

6. Analytical implication

The implication is that our next analysis should focus on high-risk patient profiles and care-transition gaps. Missing follow-up information limits visibility, while prior admissions, chronic conditions, length of stay, and age appear to be useful early risk indicators.

A natural next step is to test whether care coordination interventions reduce readmission risk for the highest-risk groups, rather than looking only at the average patient.

Source note: Results are calculated from the cleaned NHN patient-level analysis file, which combines patient readmission, financial impact, and care coordination datasets.